

Compile, or Sign

A short paper on the commoditization cascade and where it stops.

Microsoft cancelled internal Claude Code seats this November. The cost-benefit didn't clear. The same month, Uber burned its entire 2026 AI budget in four months. GitHub deprecated flat-rate Copilot. Anthropic moved to opaque weekly rate caps. OpenAI walked into a Y Combinator dinner and offered two million dollars in tokens for equity in every founder in the room.

Inside the standard frame for AI — Snowflake-class businesses, Microsoft-class moats, the trillion-dollar narratives carried by Anjney Midha's "gross underestimate" market sizing, Dario's *Machines of Loving Grace*, and Dan Sundheim's Netflix/Spotify-style defence of lab margins¹ — none of this should be happening. The buyer-side cancellations should be a buyer-education problem. The strategic asymmetry between Anthropic and OpenAI should be a duopoly differentiation play. The shift away from token meters should be a pricing experiment.

It is none of those. It is the visible symptom of a single underlying dynamic: every layer of the AI stack is racing to commoditize the layer next to it, the cascade keeps hitting a higher floor than the bull case predicted, and the floor it cannot get past is not capability or capital but *verification*.

The complement game

Joel Spolsky wrote the strategy down a quarter-century ago.² Demand for a product rises when the price of its complement falls, so firms invest in cheapening their complements. Microsoft funded the PC clone industry because cheap hardware sold Windows. IBM supported Linux because cheap operating systems sold consulting. Every layer of the AI stack is running the same play right now.

Hardware vendors push tooling that runs any model on any silicon — cheap models sell more silicon. Foundation labs subsidize first-party harnesses to capture token volume — cheap harnesses sell more tokens. Application companies pitch "replace the associate, replace the SDR, replace tier-one." The worker the app replaces is the app's complement.

The strategy memos answer the question of *who is commoditizing whom*. They do not answer the question of *where the wave stops*. The naive picture — every floor gets cheaper until everything is free — is wrong. Each commoditization wave hits a bottleneck and dissipates. The bottleneck is the layer whose complement cannot be written down. The remainder of the essay walks down the stack to find it.

Walking down the stack

Hardware takes a cost-pass-through tax. Real, durable, capped. NVIDIA and TSMC are taxing every layer above them. They will not become Microsoft-class businesses because hardware sells joules per FLOP, not outcomes.

Foundation labs appear to be the winners — eighty-percent gross margins, eight-figure enterprise contracts, valuations on platform-software comparables. But the margin is a selection artifact. Coase showed in 1972 that a monopolist of a durable good cannot extract the monopoly price; Board and Pycia rescued the result in 2014 by showing that buyers with an outside option self-select out, leaving the monopolist serving only the inelastic residual at full price.³ Open-source weights are the outside option. The 80% gross margin is the signature of selection, not of moat. As OSS closes the capability gap, the inelastic tail shrinks. Pricing power survives; revenue does not.

Orchestration — agentic harnesses, planner-worker frameworks, eval schedulers — is structurally a tax. Recent empirical data has hub-spoke architectures (planner model decomposing, worker models executing, planner recombining) losing by a factor of four to letting models compete on the same task. The orchestration overhead is itself a full planner-model run; the market wins by retry; the labs absorb orchestration as a feature in their release notes. The category survives only when it bundles a verifier into the product.

Applications appear to win where the lab cannot. But the application that resells inference at retail — Cursor, Windsurf, the horizontal AI SaaS cohort — gets squeezed between the lab's wholesale margin and the buyer's outside option. The wrapper's marginal value decays as the buyer consumes more inference. Smart apps that resell cognition make dumb businesses. The application companies that survive are not selling AI. They are selling a *verifier substrate* that happens to run AI inside it.

Halter's collars on the cattle. Waymo's sensors on the road. Abridge in the medical encounter. OpenEvidence in the chart. Sardine's loss data on the chargeback. Different industries, identical pattern: the rollout owns the substrate that produces the verification signal, owns the model that processes it, and bundles in the credentialed human signature that closes the loop. The right comparison for these businesses is Visa, the freight-rail consolidations of the 1880s, or the credit-card networks of the 1970s — capital-heavy, locally monopolistic, balance-sheet armoured. Not Snowflake.

Why the wave stops at the verifier

Why do these particular application companies win and not others? Because they own the verifier in their domain.

The verifier is the cheapest party that can decide whether AI work was done correctly. For coding, the verifier is a compiler — cheap, deterministic, productisable, anyone can run it. For an investment memo, the verifier is the partner who reads it and signs. For a pediatric intake, the verifier is the clinician with seven years of training and a malpractice premium.

Define the Kolmogorov complexity of verification, $K(\text{verify})$, as the length of the shortest program that maps outputs to yes-or-no judgments given the verifier's information set.⁴ For a compiler, $K(\text{verify})$ is small: language semantics plus a test harness. For a partner reviewing a brief, $K(\text{verify})$ is enormous: the assigned judge's history, opposing counsel's playbook, the client's risk tolerance, the firm's strategy. Most of it tacit.

The central claim of the essay is one sentence:

The rent on a task tracks $K(\text{verify})$ in the domain of redemption — not the K -complexity of the task itself, not the technical capability of the model.

This is the answer to the cycle's central puzzle. The strategy memos say "AI replaces low- K -complexity work" and stop. But the human at the bottom of the workflow keeps getting paid for what looks like trivial cognitive work. The pediatric intake takes a frontier model fifty milliseconds. The license that authorises a human to sign for it takes seven years of training and a board exam. The cognitive task got faster and cheaper. The signature did not.

Two strategies

The labs know this. Their visible behaviour is the response.

Anthropic is going vertical into regulated industries — clinical decision support, claims adjudication, regulatory affairs, structured-finance underwriting — because in those domains $K(\text{verify})$ stays high and the residual rent accrues to whoever owns the verifier substrate. Direct enterprise contracts account for ~80% of the lab's revenue. First-party harnesses (Claude Code, Coefficient) pull token volume inside the lab. The OAuth restriction on third-party harnesses pulls telemetry inside the perimeter. The bet: $K(\text{verify})$ remains heterogeneous, the substrate stays scarce.

OpenAI is going horizontal — multi-vendor supply agreements with Microsoft, Amazon, and Google flatten hyperscaler dependence; the \$2M YC token offer buys equity in tomorrow's app businesses paid in OpenAI's own currency; consumer products extend perimeter into consumer regions. The bet: $K(\text{verify})$ falls fast, the substrate productises across domains, volume across many small substrates dominates depth in any one.

Two beliefs about the same parameter. Two corner solutions to the same optimisation. The labs are not competing for the same customer. They are betting on opposite futures of $K(\text{verify})$.

Bank runs

The strategic stress is now visible at single-buyer scale. Microsoft cancelling Claude Code is the equivalent of a depositor at a Free Banking-era bank showing up for specie and being told the redemption doesn't clear.⁵ The bank survives the single withdrawal. The information about whether the bank could honour at face propagates. Other large enterprises run their own cost-benefit against the same flat-rate subscription. Stablecoin history is the modern training set: TerraUSD destroyed \$45B in a week, USDC depegged in March 2023, Tether's reserves were 27.6% of issued tokens for two and a half years.⁶ Private currencies without sufficient backing fail not gradually but suddenly, when buyer confidence cracks.

Anthropic's weekly rate caps and OpenAI's priority-service contracts are tacit admissions that the per-token meter is not a stable medium of exchange. The labs are converting their banknotes into bilateral capacity reservations — the retreat private banks made when they ran out of specie.

Three resolution paths. State imposition (a National Banking Acts analog under which USG declares the inference economy strategic and absorbs the labs into a regulated-issuer framework).⁷ Serial failure (smaller labs absorbed, larger labs compressing to defence-contractor multiples). Multipolar fragmentation (US, Chinese, European parallel systems with different verifier-credentialing regimes). Probability is roughly equal across the three on a five-year horizon. The realised outcome will be a combination.

What this implies

What the cycle has gotten wrong is the model class. Lab equity is priced on platform-software comparables. The realised revenue and cost structure look like cost-pass-through infrastructure. The implied multiple compression is the modal expected outcome, not a tail risk.

The institutions that benefit are verifier-substrate owners: credentialing bodies, audit firms, vertical software companies that own the substrate in their domain. Underwriters Laboratories, the AICPA, Moody's, and their international counterparts on the institutional side; Halter, Waymo, Abridge, OpenEvidence, Sardine on the cybernetic-rollup side. They trade at infrastructure multiples and they earn them. Compute and energy infrastructure benefits regardless of which resolution path obtains — the trillion-dollar cluster buildouts happen on every branch of the tree.

The complement game has a quiet last move. Every layer in a stack races to commoditize its complement. The layer that survives is the one whose complement is structurally un-commoditizable. Hardware survives because joules don't compress. Humans at the edge survive because the meter has to live on a signature, and signatures live on people.

The credentialed human at the bottom of the signature chain is the closest thing the AI cycle has produced to specie reserves. She holds value across every redemption scenario. No lab can capture her without integrating into the substrate where her signature lives. The wage of the doctor, the lawyer, the engineer, the underwriter, the pilot rises in real terms over the next five years. The middle of the labour market — the analyst, the coordinator, the entry-level associate, the role whose function was to assemble inputs for someone else's signature — compresses out of every column simultaneously. The political debate about AI and labour has been sighting the wrong tier.

The cycle has been pricing the wrong substrate. The repricing comes through state intervention, through consolidation, or through fragmentation. All three are visible in the current data.

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1. Anjney Midha (Anthropic seed investor, ex-a16z): "in the early days, a couple Anthropic cofounders and I started doing weekly working sessions for our first fundraise we needed a coherent view of our market size and arrived at an answer that felt reasonable: ~\$800B/year as Andrej has discovered, that was a gross underestimate" — X.com, Oct 2025. Dwarkesh Patel: founders and early investors of frontier labs will "own a galaxy." Dario Amodei, "Machines of Loving Grace" (Oct 2024): trillion-dollar revenues by 2030, 20% annual GDP growth in the developing world, AI-replicated East Asian-style ~10% growth. Dan Sundheim (D1 Capital), Invest Like the Best Ep. 460 (Feb 2026): LLM businesses combine Netflix-style fixed-asset economics with Spotify-style personalisation stickiness; the commoditisation thesis "is more or less irrelevant."
 2. Joel Spolsky, "Strategy Letter V: Commoditize Your Complement", joelonsoftware.com (June 2002). The framing borrowed from undergraduate microeconomics: "Demand for a product increases when the prices of its complements decrease." Hamilton Helmer, *Seven Powers* (2016), on counter-positioning: the labs' token meter is counter-positioned against the SaaS flat-rate model, and SaaS cannot retrofit per-token pricing without breaking its customer relationship.
 3. Ronald Coase, "Durability and Monopoly," *Journal of Law and Economics* 15 (1972). Board & Pycia, "Outside Options and the Failure of the Coase Conjecture," *American Economic Review* 104:2 (Feb 2014). The labs' eighty-percent gross margins are the Board–Pycia selection signature, not a moat.
 4. The K(verify) construction extends conditional Kolmogorov complexity (Li & Vitányi, *An Introduction to Kolmogorov Complexity*, 4th ed., Springer 2019) using a verifier's tacit information set as the conditioning variable. Underlying economic machinery: Akerlof, "Lemons" (*QJE* 1970); Holmström, "Moral Hazard and Observability" (*Bell J. Econ.* 1979); Bergemann & Bonatti, "Markets for Information" (*Annu. Rev. Econ.* 2019).
 5. American Free Banking Era (1837–1863): state-chartered commercial banks issued their own banknotes, redeemable on demand for specie. Notes circulated at face inside the issuing bank's home county and discounted with distance. The resolution was the National Banking Acts of 1863–1864 — unified federal currency, taxes that killed state banknotes, federally chartered banks. See Hugh Rockoff, "The Free Banking Era: A Reexamination," *Journal of Money, Credit and Banking* 6:2 (1974), pp. 141–167; Calomiris & Haber, *Fragile by Design* (Princeton, 2014).
 6. TerraUSD / LUNA collapse, May 2022 — algorithmic peg failed; ~\$45B value destroyed in a week. USDC depeg, March 2023 — Silicon Valley Bank held Circle's reserves and failed. Tether reserve fraction during 2016–18 was 27.6% per the NY Attorney General's 2021 settlement. Three independent demonstrations that private currencies without sufficient backing fail discontinuously when confidence cracks.
 7. Leopold Aschenbrenner, "Situational Awareness: The Decade Ahead" (June 2024), particularly the "The Project" section: predicts USG absorption of frontier labs into a Manhattan-Project-style consortium by 2027–28 under Lockheed/Boeing-type defence-contracting arrangements, triggered by capability advancement and US-China geopolitical pressure.